

# Abstract Linear Algebra

Prof. Lanier

[jlancier8@uchicago.edu](mailto:jlancier8@uchicago.edu)

Please pick up a copy of the handouts.

Try those warm-up problems.

For you to do ASAP:

- Finish read sections 1-4 of Chapter 1 of *LADW*
- Come to office hours! Today 3:30-4:30pm
- Get started on HW1
- Flex planning due Sunday

# Today in class:

- Explain homework procedures
- Discuss warm-ups
- Overview content of chapters 3 and beyond
- Discuss formal definitions of groups, fields, and vector spaces

## Three parts to homework:

### **Written: Due Fridays at 11:59pm**

Submit 8 problems through Canvas, 4 for correctness and 4 for completion

Problems are from the sections of *LADW* for the week, plus 5 more

Stretch yourself!

I give some suggestions for problems from *LADW*

Mark problems you want graded for correctness with “G”.

Mark completion problems you want feedback on with “F”.

Exemplary solutions will be selected each week; flex points & shared with class

Small flex incentive to LaTeX newcomers to learn and use it for some HWs

### **Homework reflection and response: Due Fridays at 11:59pm**

Chance to think over the homework and consider what you’ve learned

Request solutions for specific problems

### **WeBWorK: Due Saturdays at 11:59pm**

10 problems

Your login will be in your Canvas inbox today.

Your initial password is the same as your login.

Instructions for using WeBWorK are on Canvas.

Let me know if you have any issues with logging in or using the site.

## Warm-up 2

Compute the following real matrix-vector product:

$$\begin{pmatrix} 8 & -2 & 4 & 23 & 6 \\ 0 & 4 & 31 & 2 & 8 \\ -5 & 4 & -2 & 4 & 20 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} =$$

What are the domain and codomain of the linear transformation defined by the matrix?

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$$\mathbb{R}^5 \rightarrow \mathbb{R}^4$$

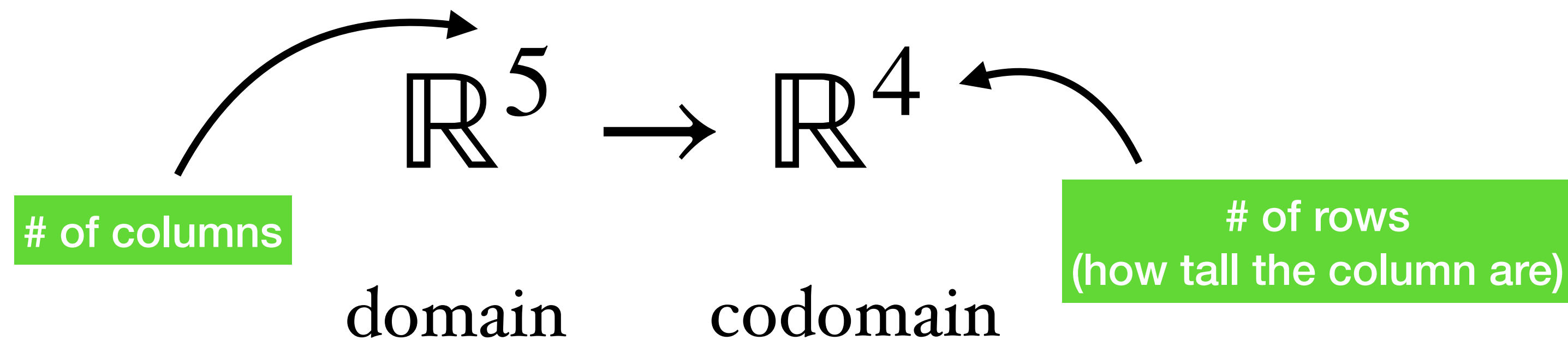
domain      codomain

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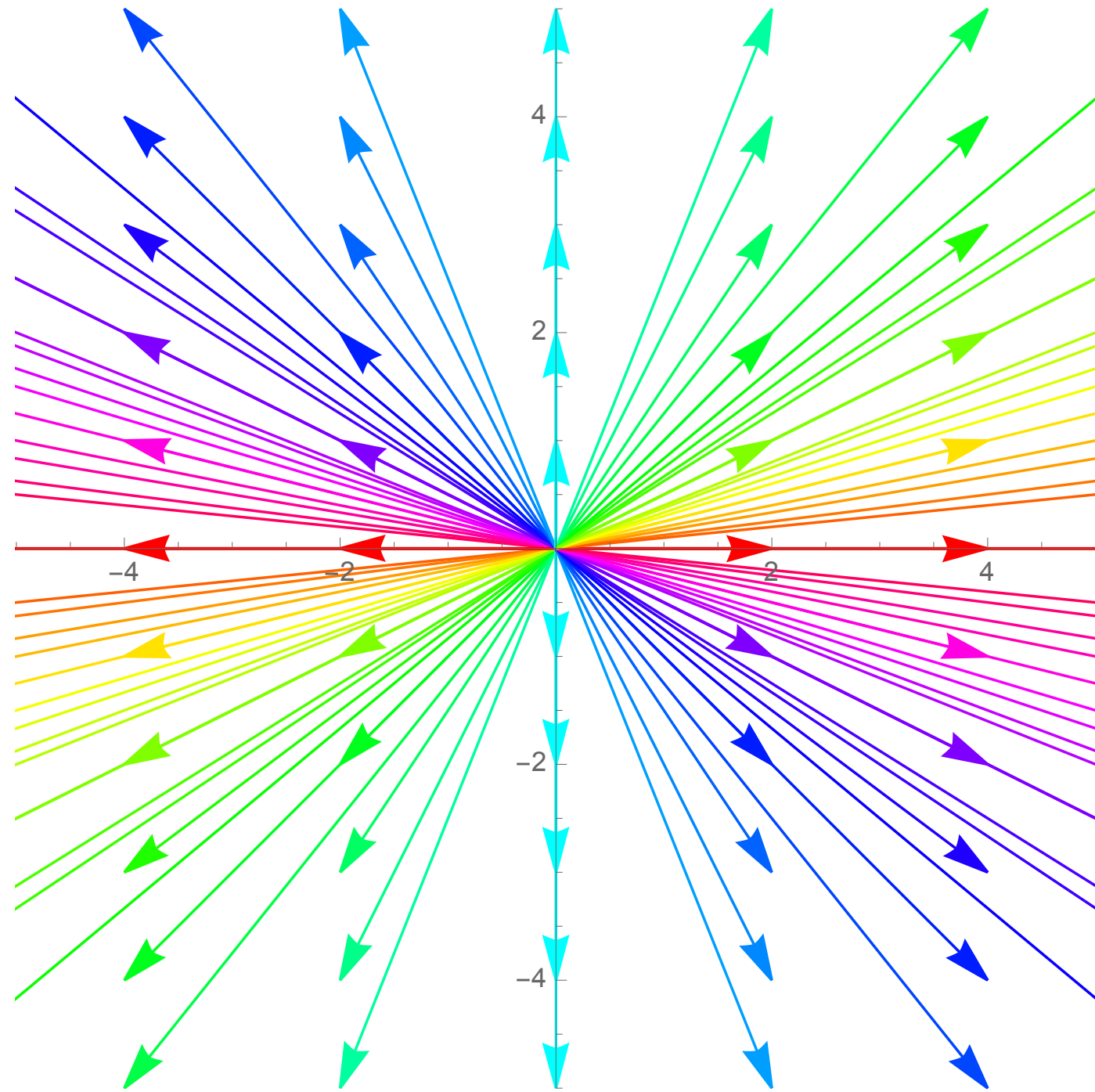
- Ch 1: vector spaces, bases, linear transformations, matrices
- Ch 2: solving linear systems, row reduction, inverse matrices
- Ch 3: determinants
- Ch 4: eigenvalues and eigenvectors
- Ch 5: inner products, orthogonality
- Ch 6 & 9: matrix decompositions

## Ch 3

determinant: “global stretch factor” of a linear transformation  
 $\det(T)$  is the factor by which  $T$  scales area (or  $n$ -dimensional volume).

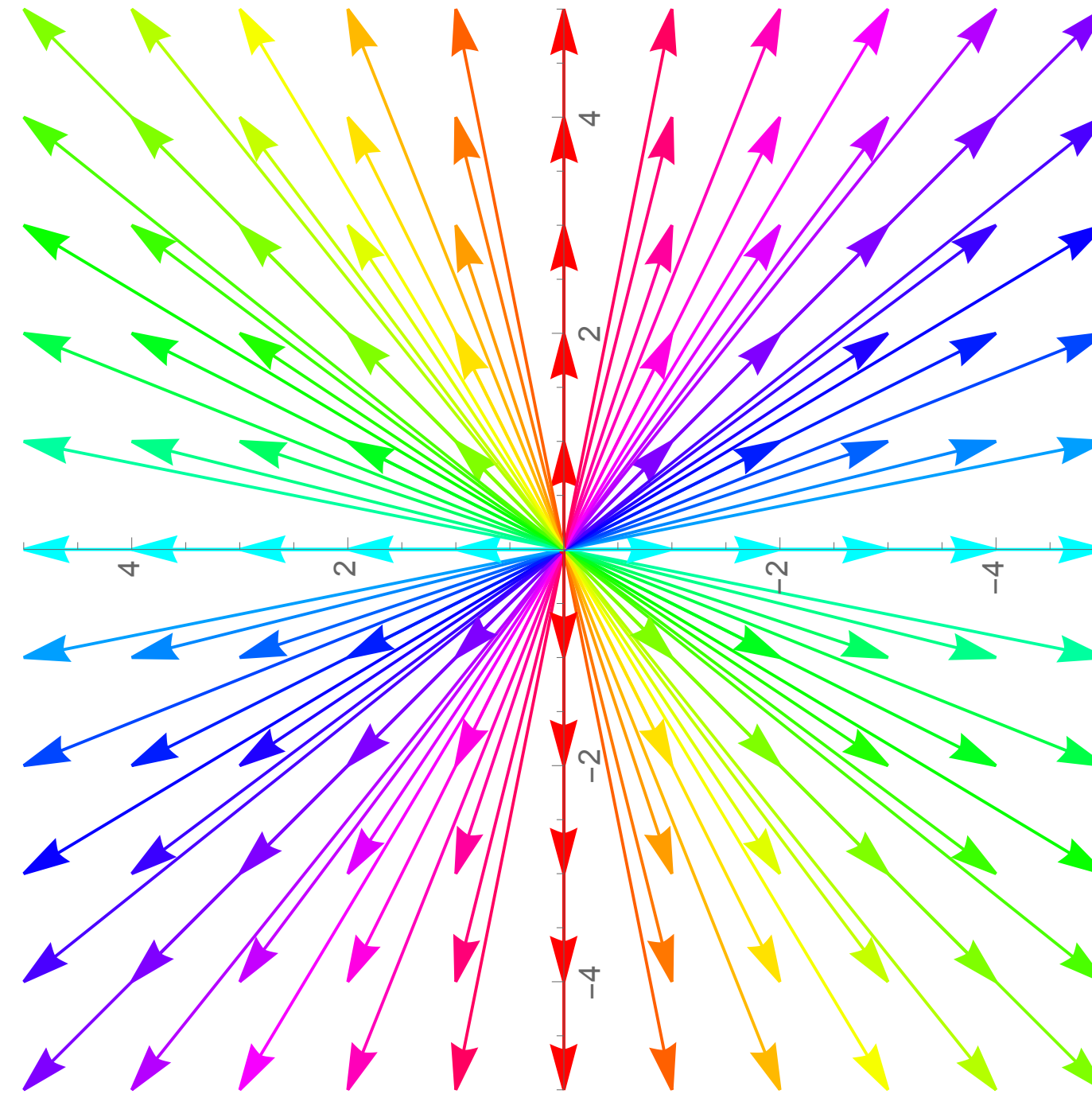
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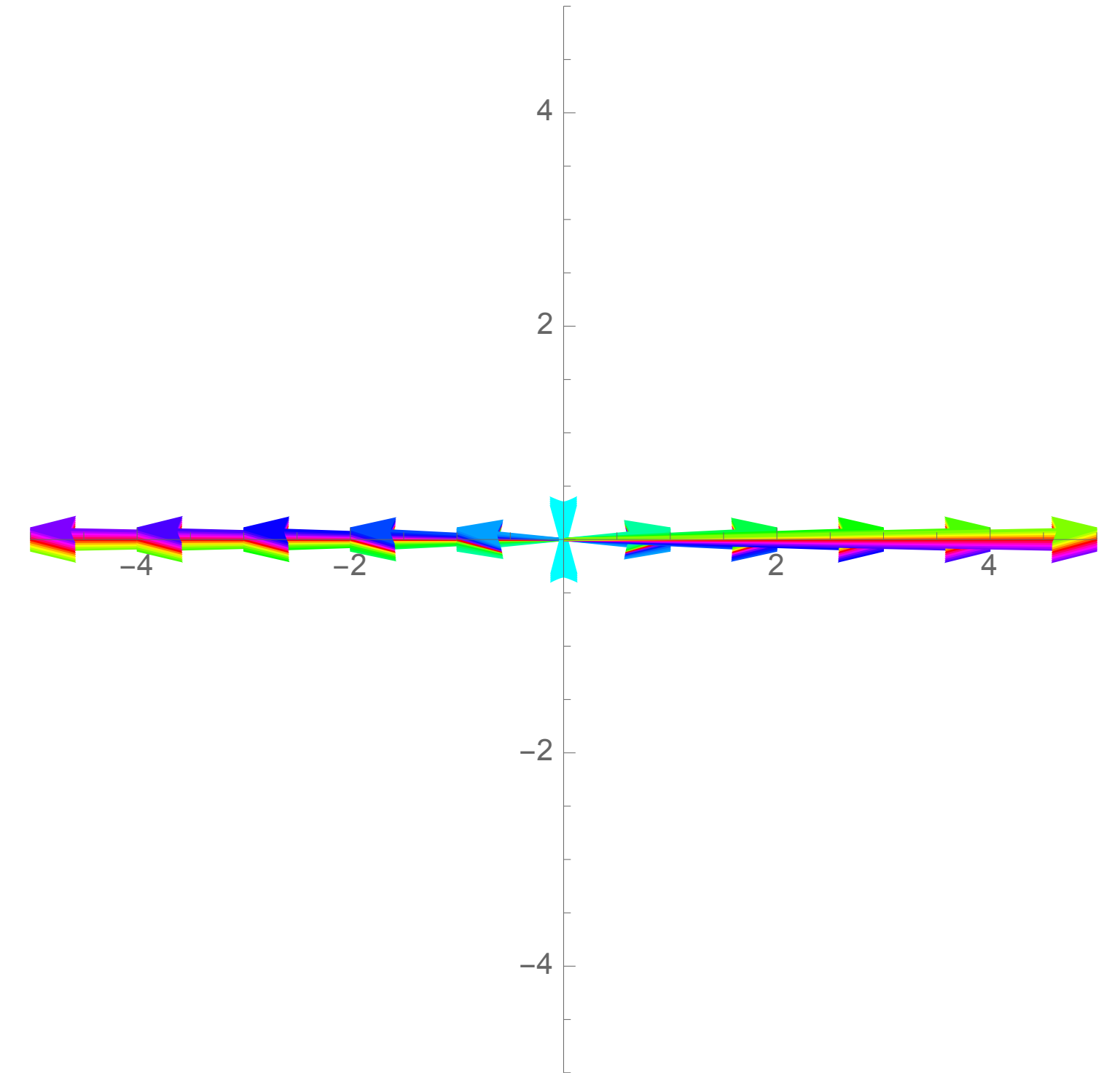
$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal



$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

rotation

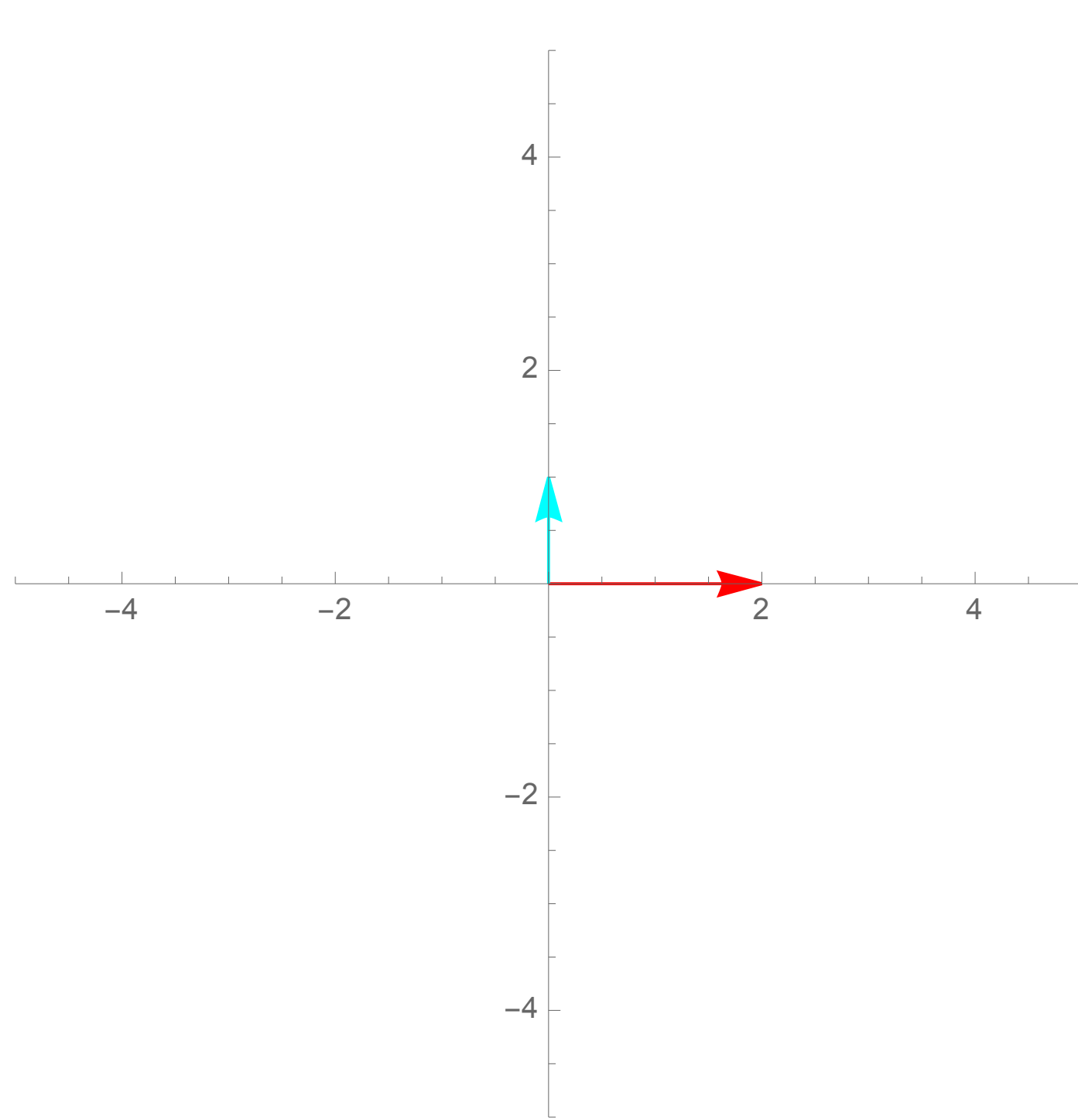


$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

projection

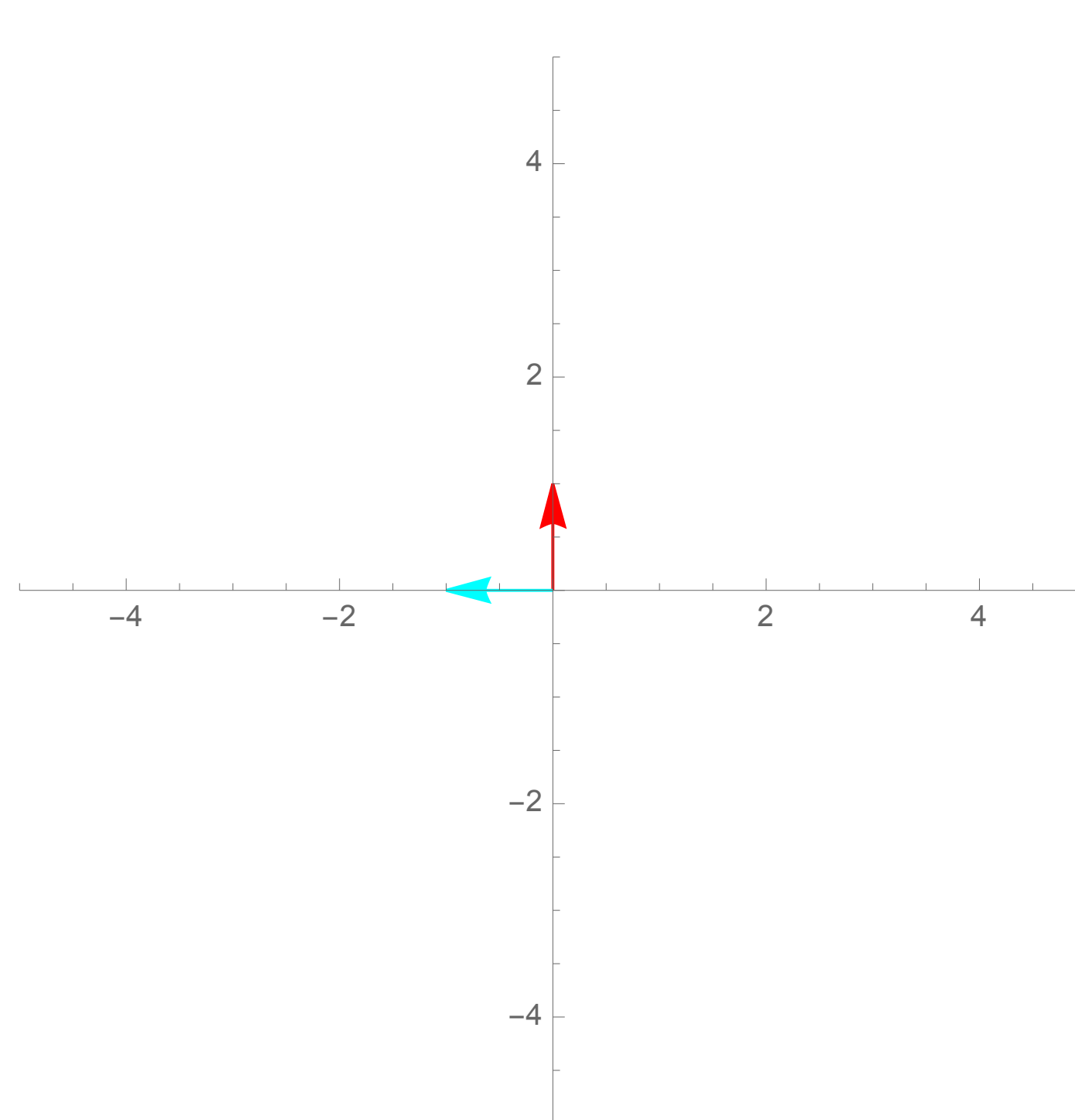
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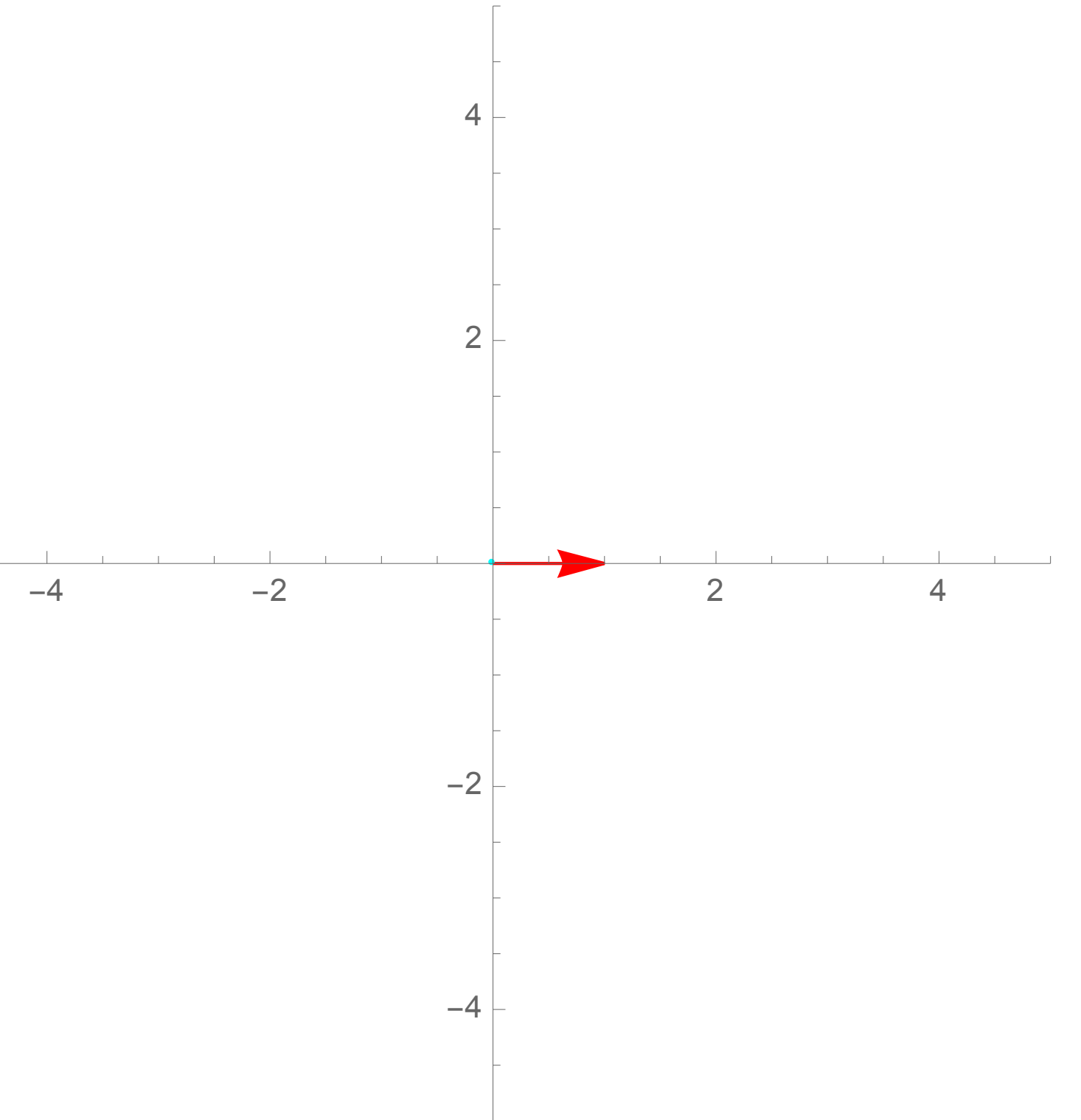
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rotation

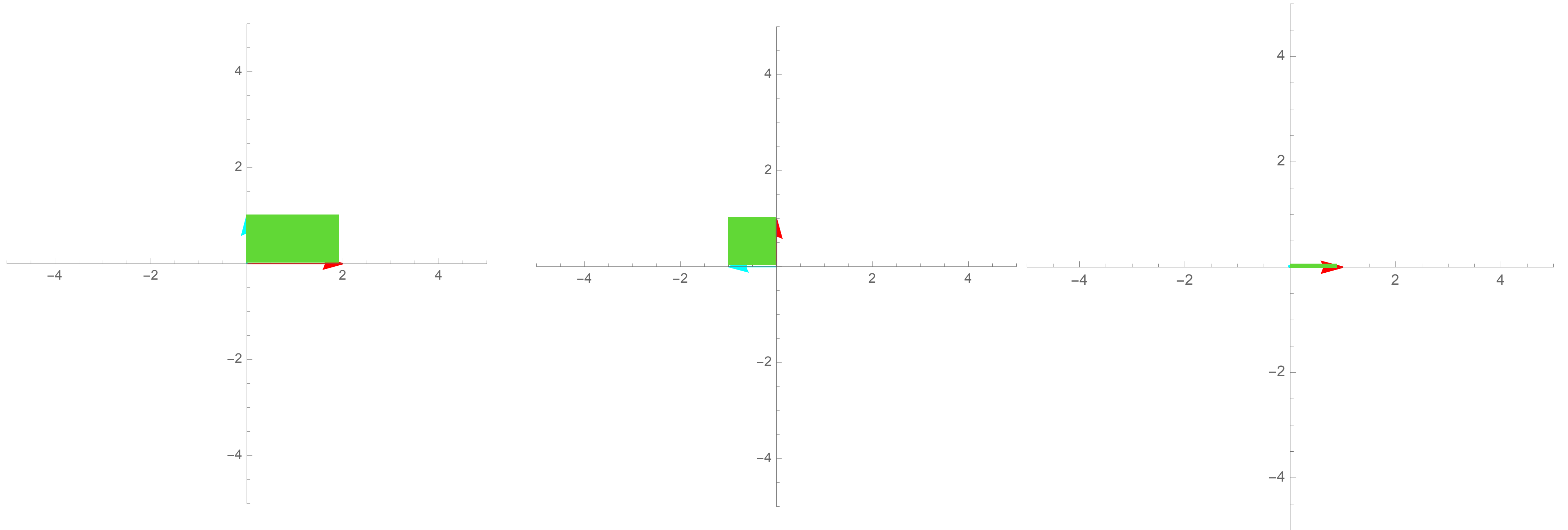


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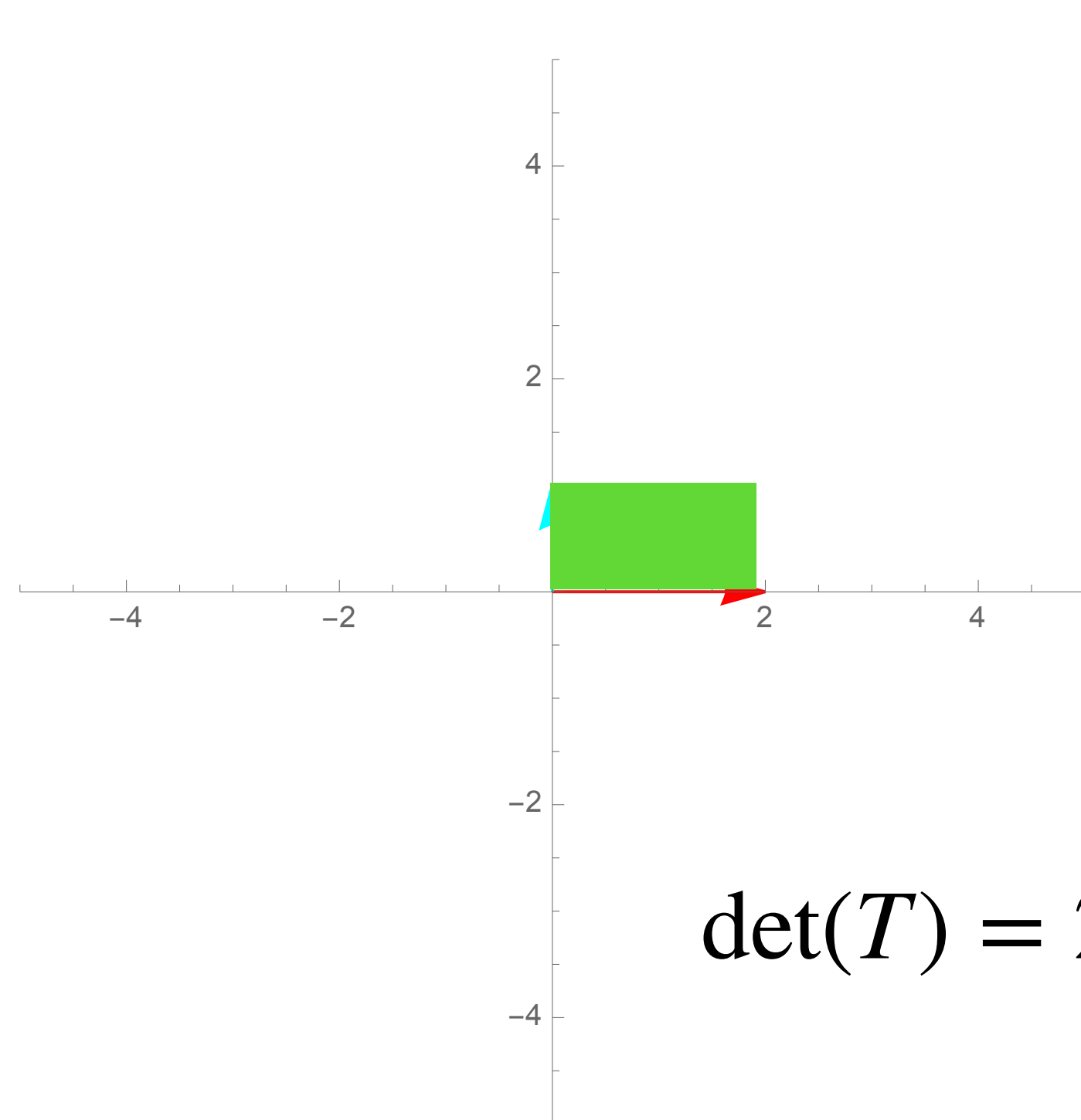
rotation

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projection

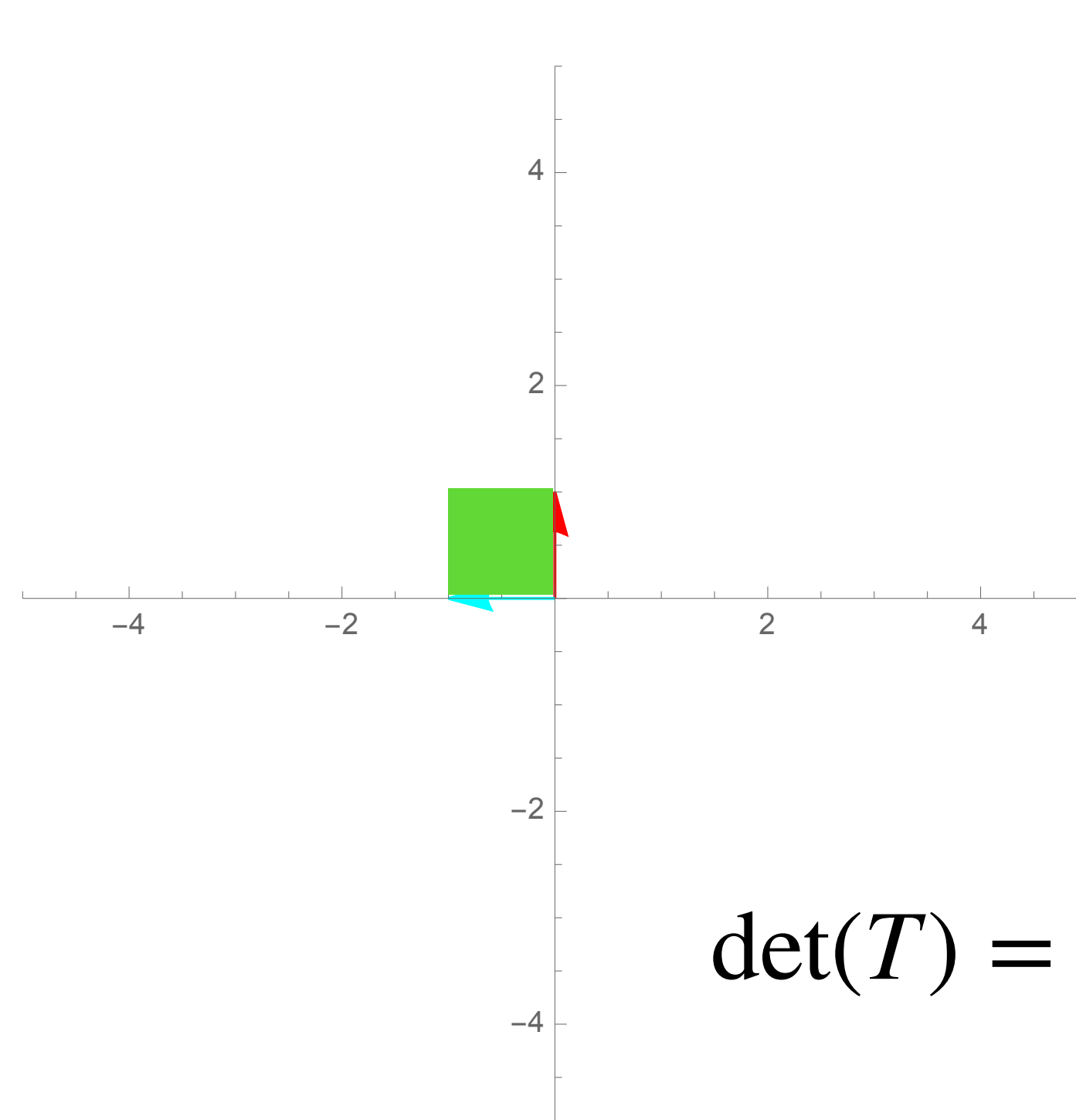
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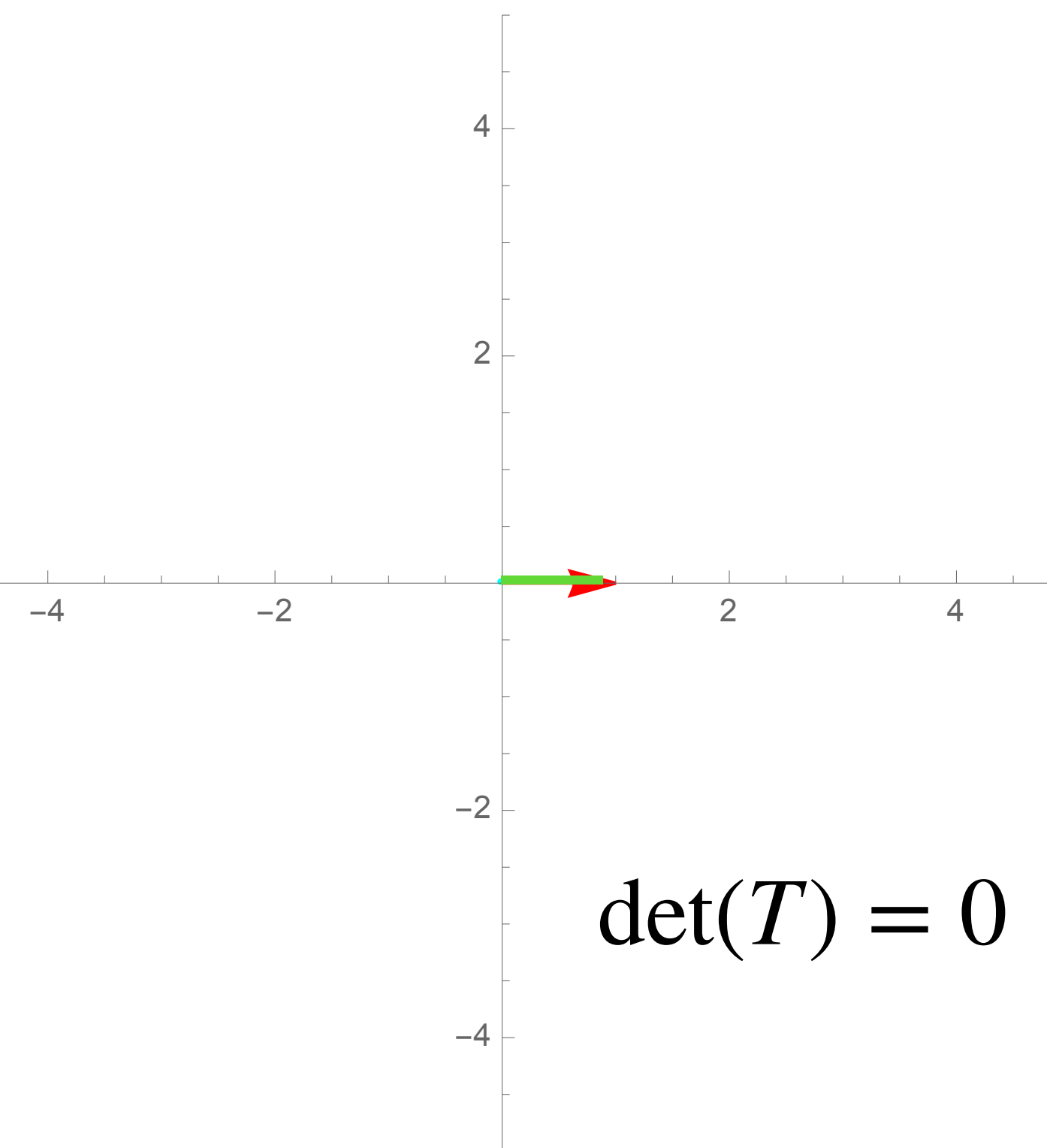
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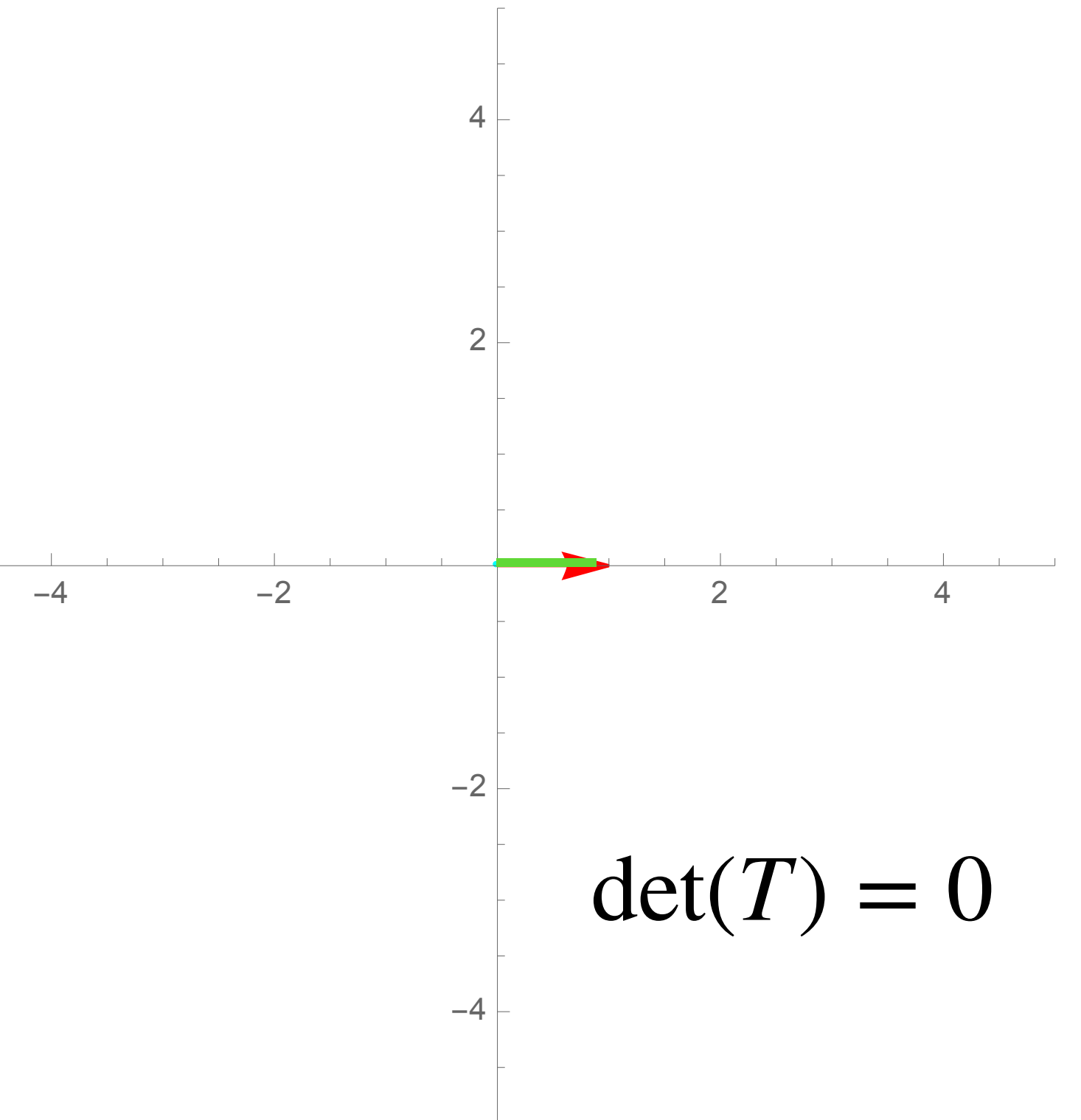
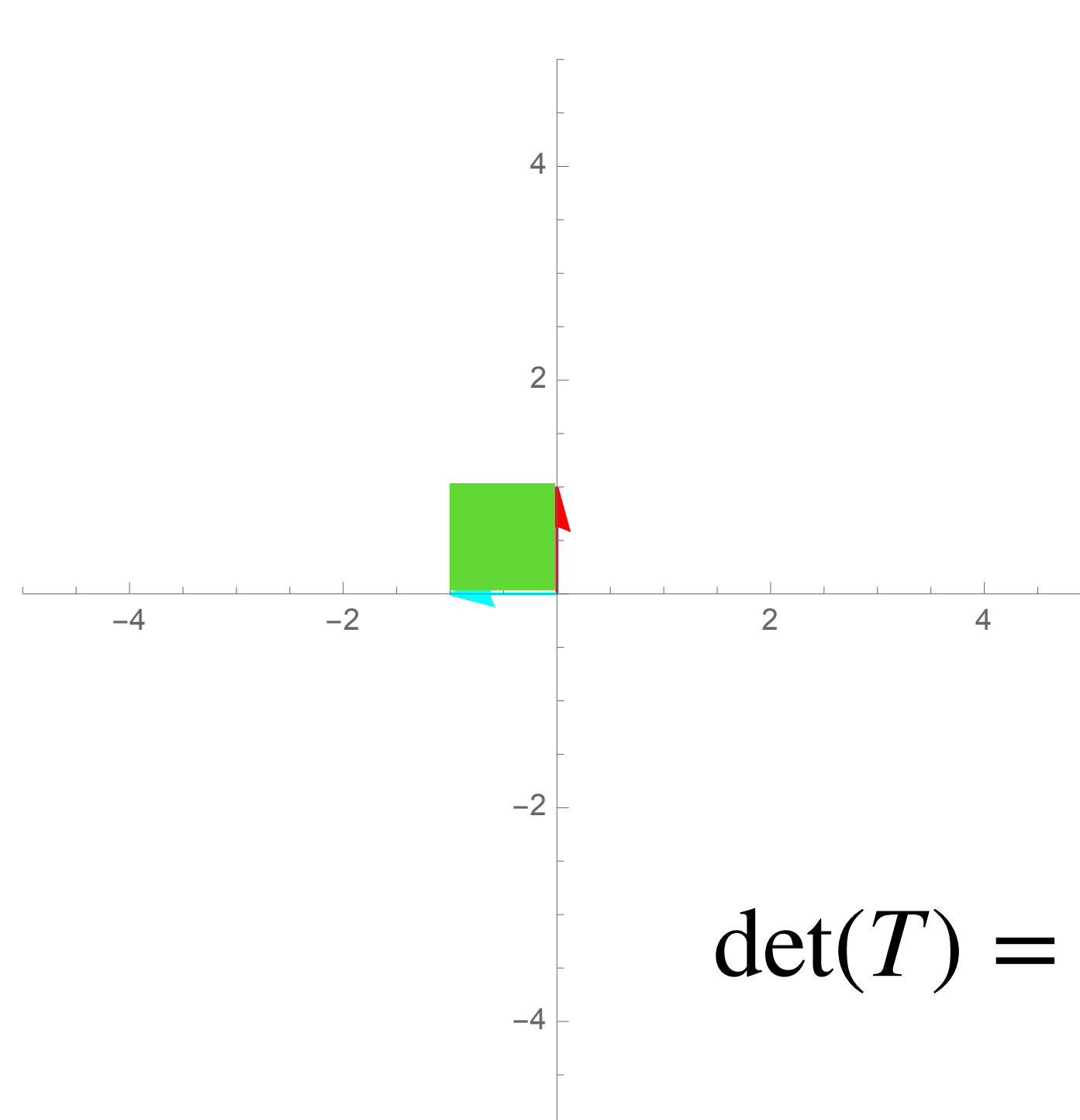
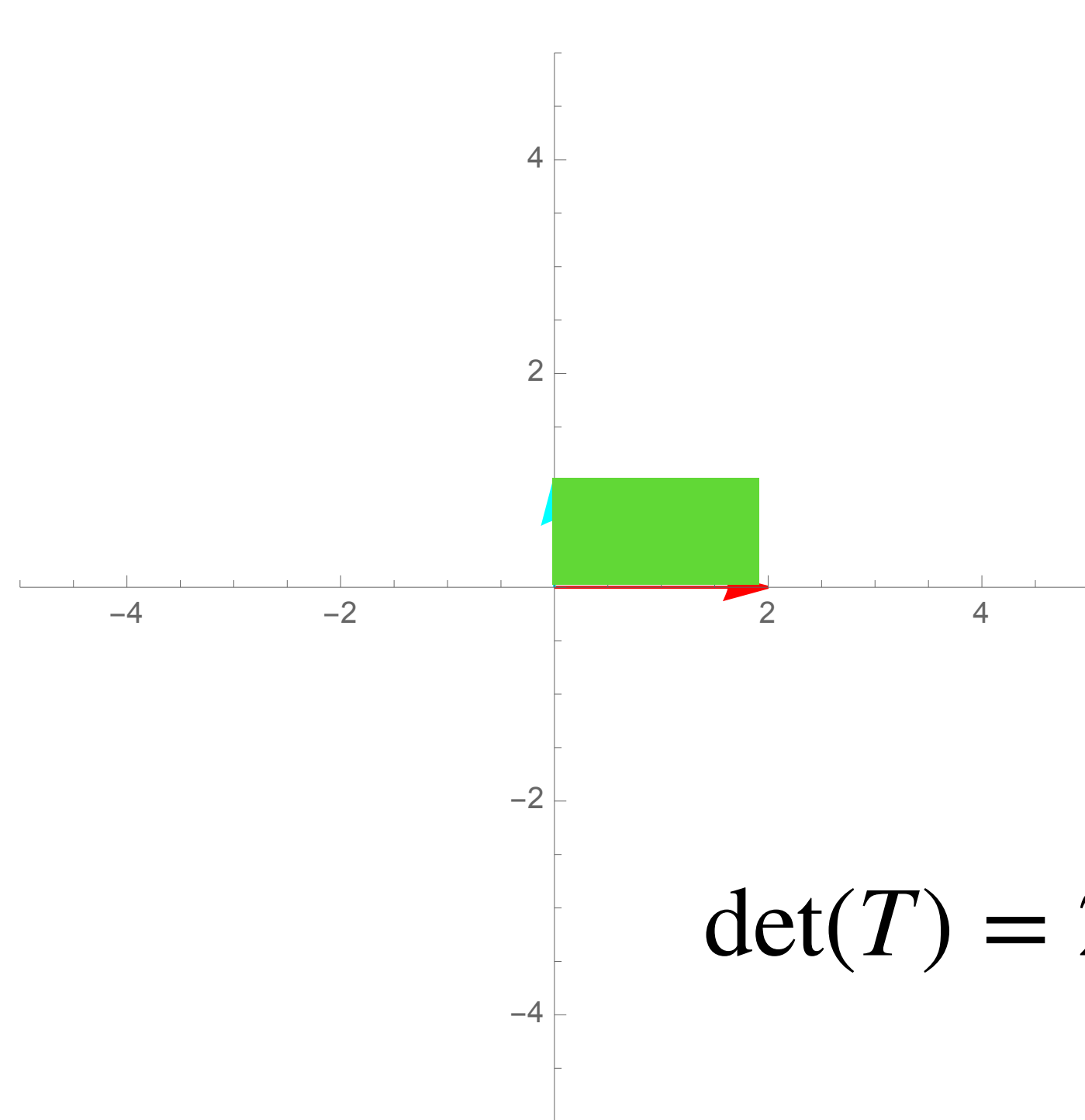


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$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

diagonal

How can we calculate  $\det(T)$ ?

What info does  $\det(T)$  give us about  $T$ ?

What properties do/can/must  $\det(T)$  have?

## Ch 4

eigenvalues: “local stretch factors” of a linear transformation

eigenvectors: vectors that a linear transformation acts upon by scaling

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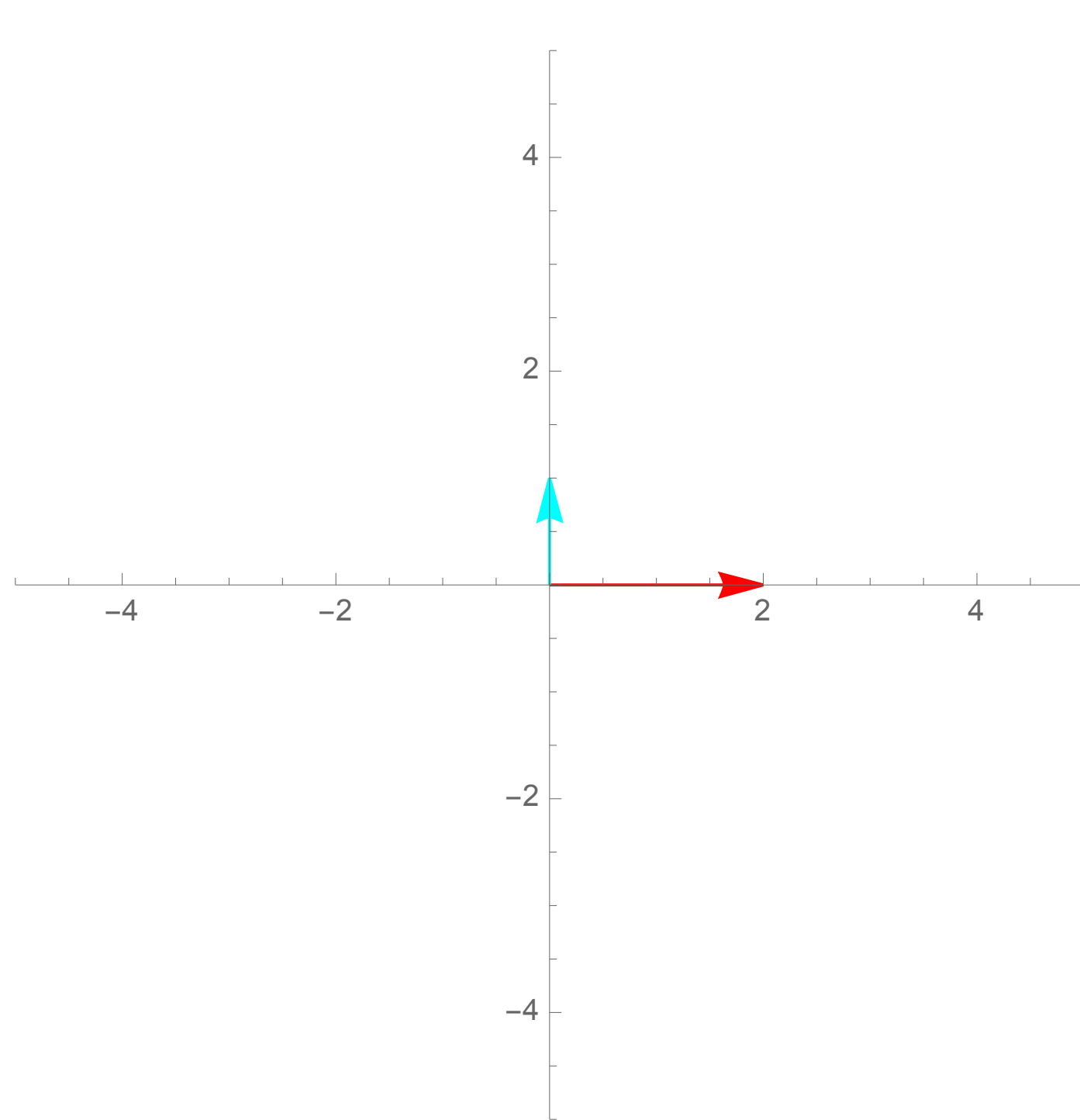
eigenvectors: vectors that a linear transformation acts upon by scaling

“eigen” is German and means something like  
“belonging to” or “characteristic”

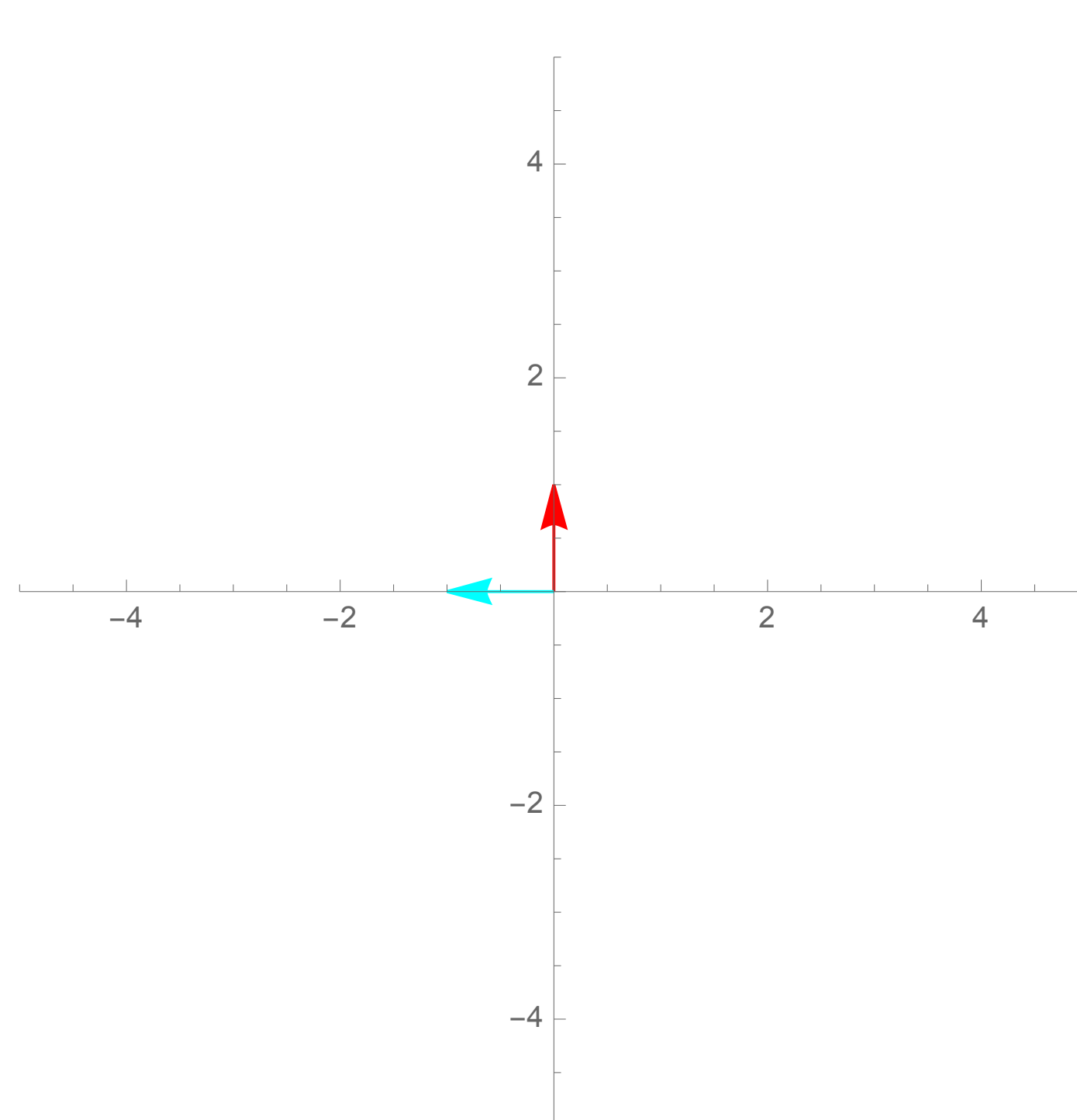
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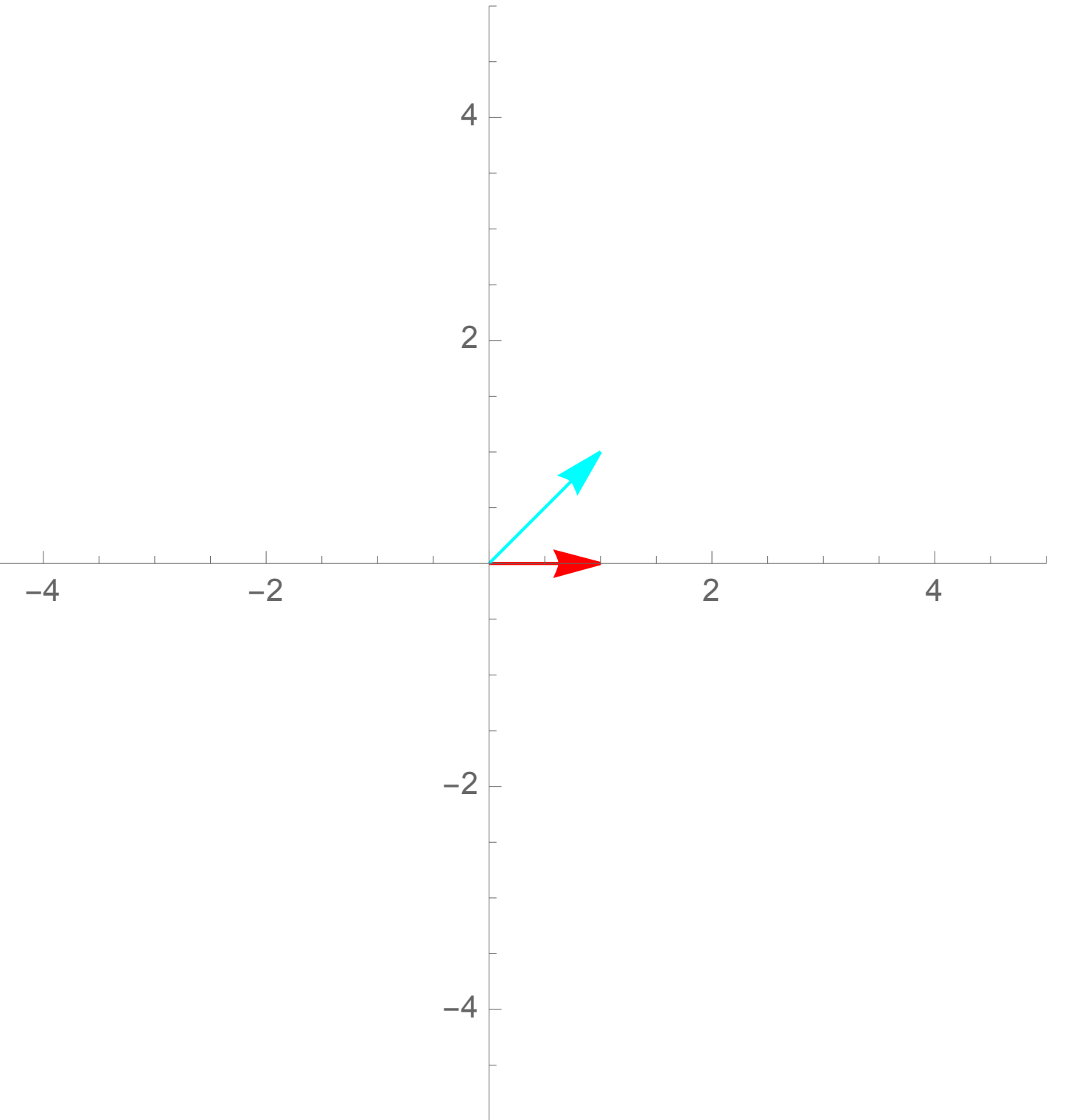


$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

rotation

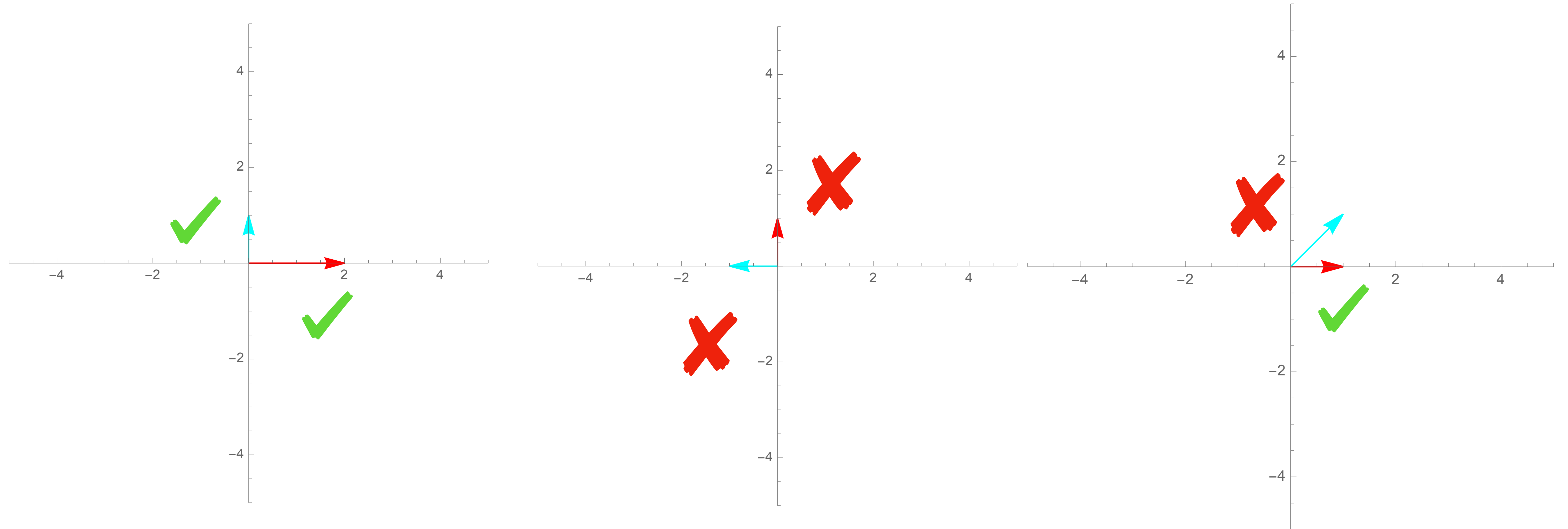


$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

shear

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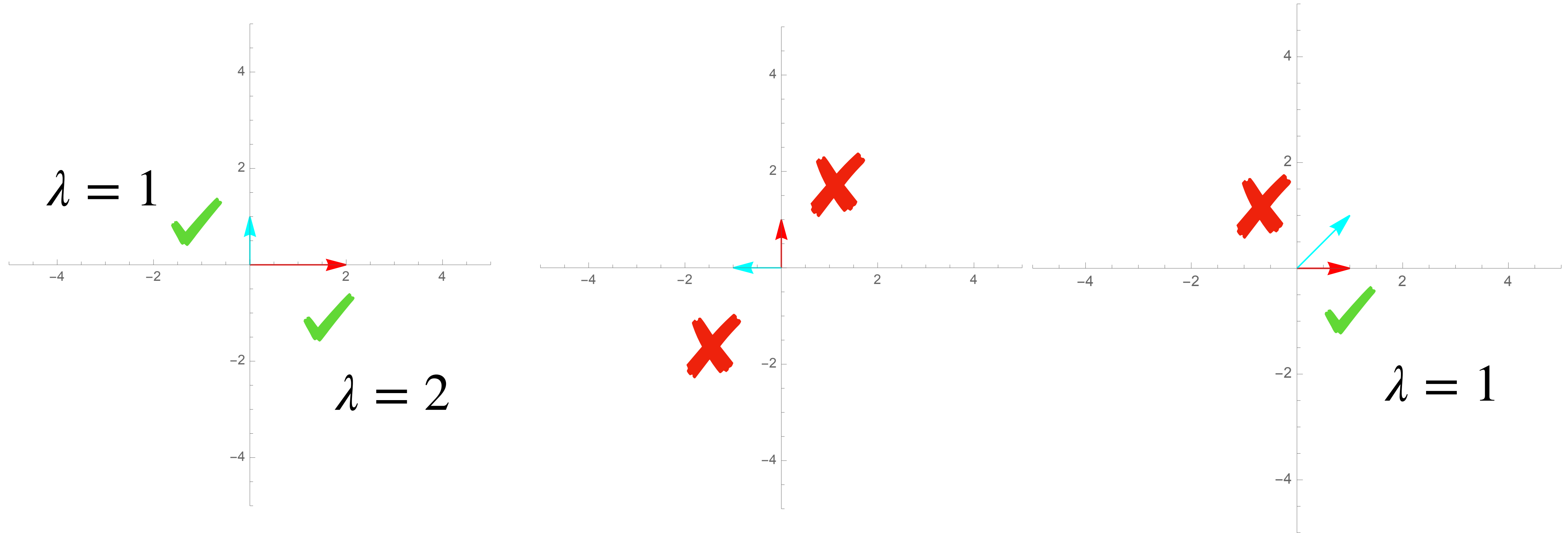
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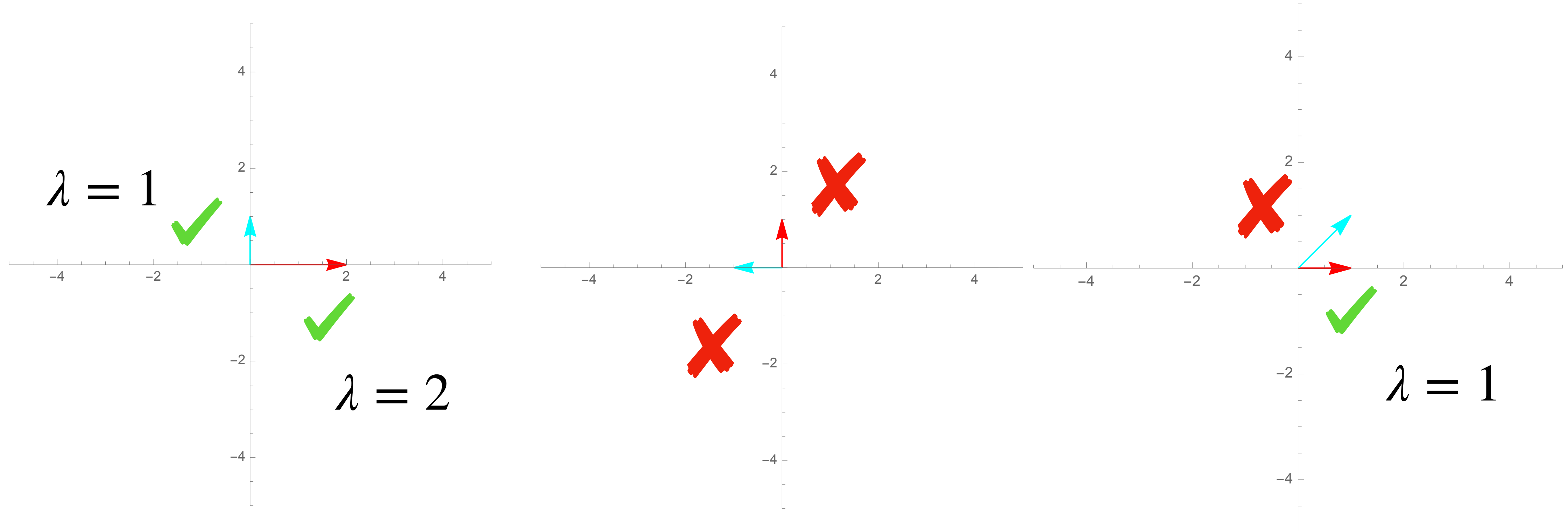
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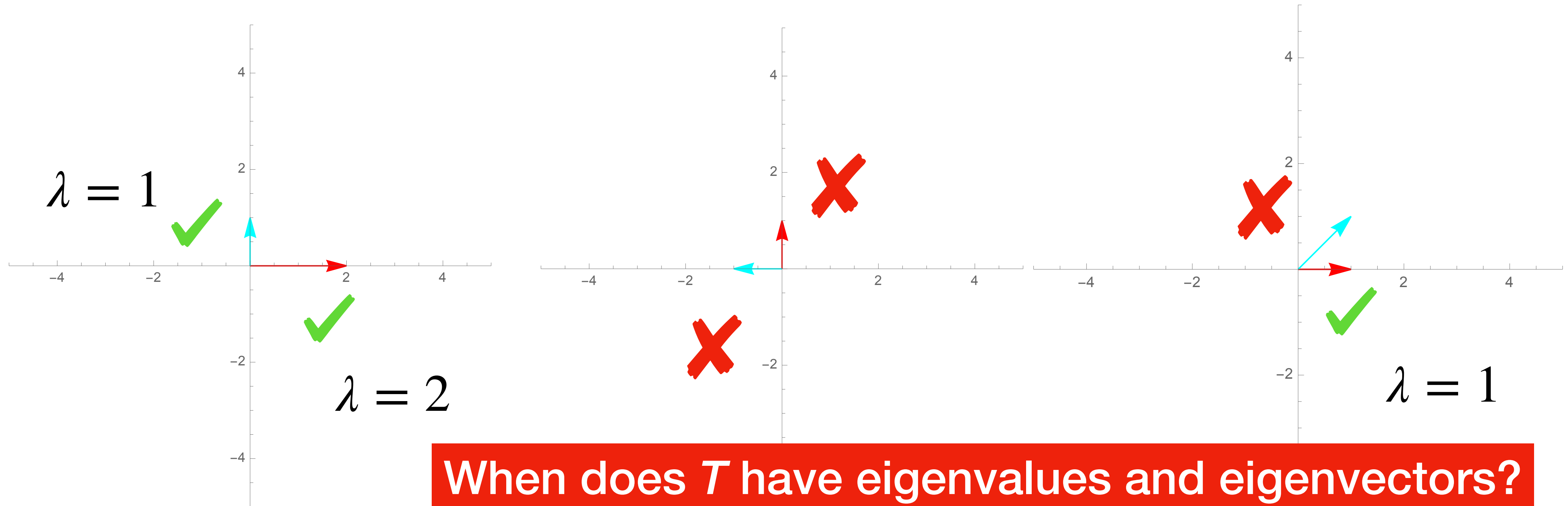
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Note: eigenvectors don't have to be standard basis vectors.

# Ch 4

eigenvalues: “local stretch factors” of a linear transformation  
eigenvectors: vectors that a linear transformation acts upon by scaling



$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

When does  $T$  have eigenvalues and eigenvectors?

When it does, how can we find them?

What info do they give us about  $T$ ?

What properties do/can/must they have?

# Ch 5

## inner products and orthogonality



These pairs of vectors are indistinguishable in  $\mathbb{R}^2$ ,  
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But by layering on additional structure—an inner product—  
we can measure angles between vectors.

e.g. the dot product

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But by layering on additional structure—an inner product—  
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We can also study linear transformations  
that preserve an inner product.

Chs 6&9

matrix decompositions

Chs 6&9

**number**decompositions

Chs 6&9

**number** decompositions

$$1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$$

Chs 6&9

**number** decompositions

$$1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$$

10

$$1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$$

$$10$$

$$2 \times 5$$

$$1^2 + 3^2$$

prime, composite

square

# Chs 6&9

## number decompositions

$$1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$$

$$10$$

$$2 \times 5$$

$$1^2 + 3^2$$

prime, composite  
square

Fundamental theorem  
of arithmetic:

Every natural number has a unique  
factorization into primes, up to ordering.

Theorem (Lagrange):

Every natural number can be  
written as the sum of (at most)  
four squares.

Chs 6&9

matrix decompositions

# Chs 6&9

matrix decompositions

$$T(av + bw) = aT(v) + bT(w)$$

# Chs 6&9

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$$\begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$$

rotation, projection  
scaling

# Chs 6&9

## matrix decompositions

singular value decomposition:  
For any real square matrix  $M$   
with positive determinant, we have

$$M = QDP$$

rotation      diagonal      rotation

The diagram illustrates the singular value decomposition  $M = QDP$ . Three blue arrows point from the labels 'rotation', 'diagonal', and 'rotation' to the matrices  $Q$ ,  $D$ , and  $P$  respectively. The label 'rotation' is positioned below the arrow pointing to  $Q$ , 'diagonal' is below the arrow pointing to  $D$ , and 'rotation' is below the arrow pointing to  $P$ .

# Chs 6&9

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Jordan canonical form:  
For any complex square matrix  $A$ , we can change coordinates  
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to get a canonical  $J$  with an easy-to-understand structure

$$A = \begin{bmatrix} 5 & 4 & 2 & 1 \\ 0 & 1 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & -1 & 2 \end{bmatrix}$$

# Chs 6&9

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$$J = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

vector space:

vectors you can add and subtract with each other,  
a zero vector, inverse (opposite) vectors;  
and a way to scale the vectors that “respects” the addition

a group of vectors and a field of scalars  
and distribution

## vector space definition:

A set of vectors  $V$  and a field of scalars  $\mathbb{F}$  with operations addition and scaling such that

1. Commutativity:  $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$  for all  $\mathbf{v}, \mathbf{w} \in V$ ;
2. Associativity:  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$  for all  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ ;
3. Zero vector: there exists a special vector, denoted by  $\mathbf{0}$  such that  $\mathbf{v} + \mathbf{0} = \mathbf{v}$  for all  $\mathbf{v} \in V$ ;
4. Additive inverse: For every vector  $\mathbf{v} \in V$  there exists a vector  $\mathbf{w} \in V$  such that  $\mathbf{v} + \mathbf{w} = \mathbf{0}$ . Such additive inverse is usually denoted as  $-\mathbf{v}$ ;

*The next two properties concern multiplication:*

5. Multiplicative identity:  $1\mathbf{v} = \mathbf{v}$  for all  $\mathbf{v} \in V$ ;
6. Multiplicative associativity:  $(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v})$  for all  $\mathbf{v} \in V$  and all scalars  $\alpha, \beta$ ;

*And finally, two distributive properties, which connect multiplication and addition:*

7.  $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$  for all  $\mathbf{u}, \mathbf{v} \in V$  and all scalars  $\alpha$ ;
8.  $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$  for all  $\mathbf{v} \in V$  and all scalars  $\alpha, \beta$ .

vector space definition:

field

A set of vectors  $V$  and a field of scalars  $\mathbb{F}$  with operations addition and scaling such that

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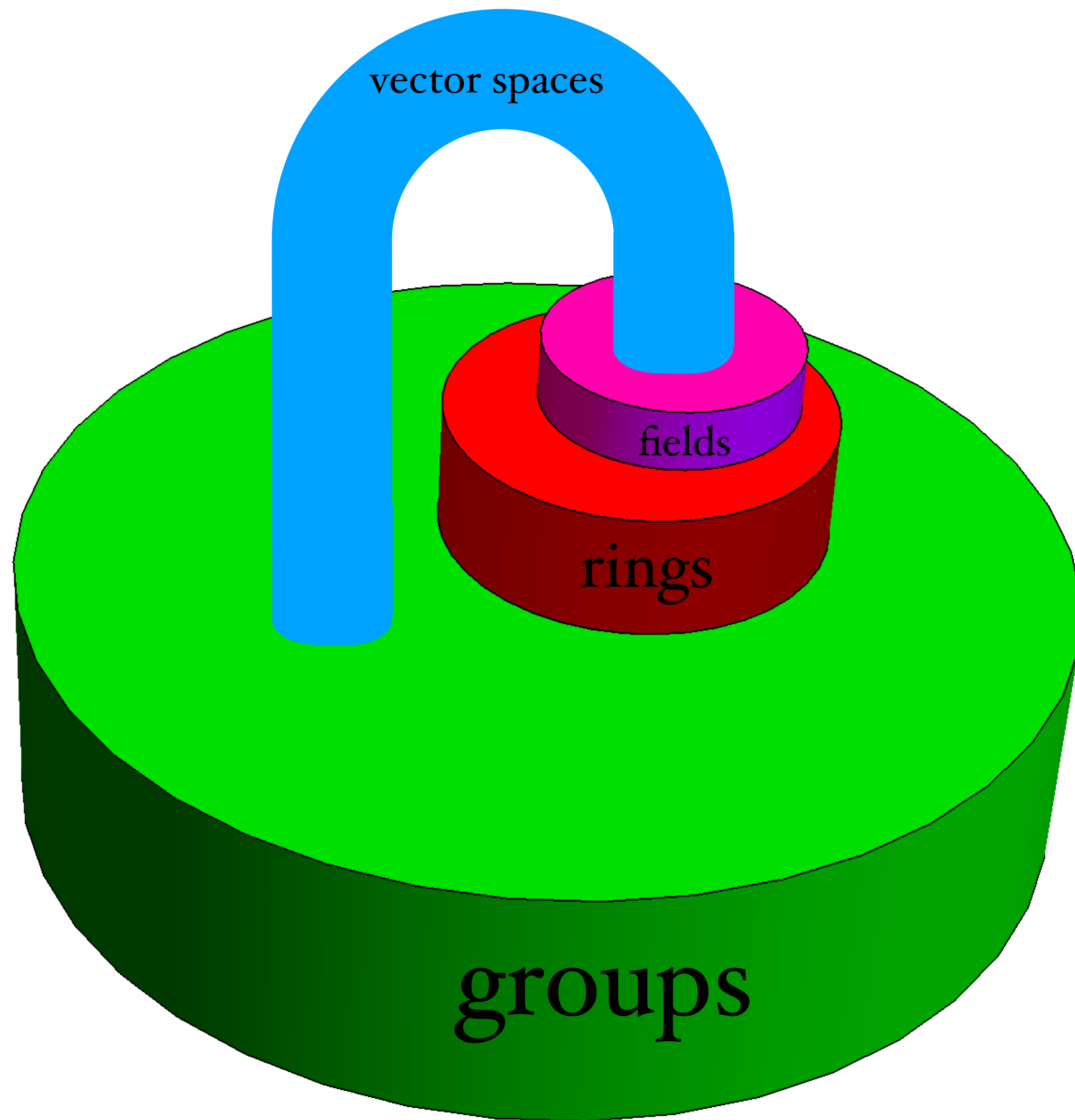
8.  $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$  for all  $\mathbf{v} \in V$  and all scalars  $\alpha, \beta$ .

commutative

group

scaling

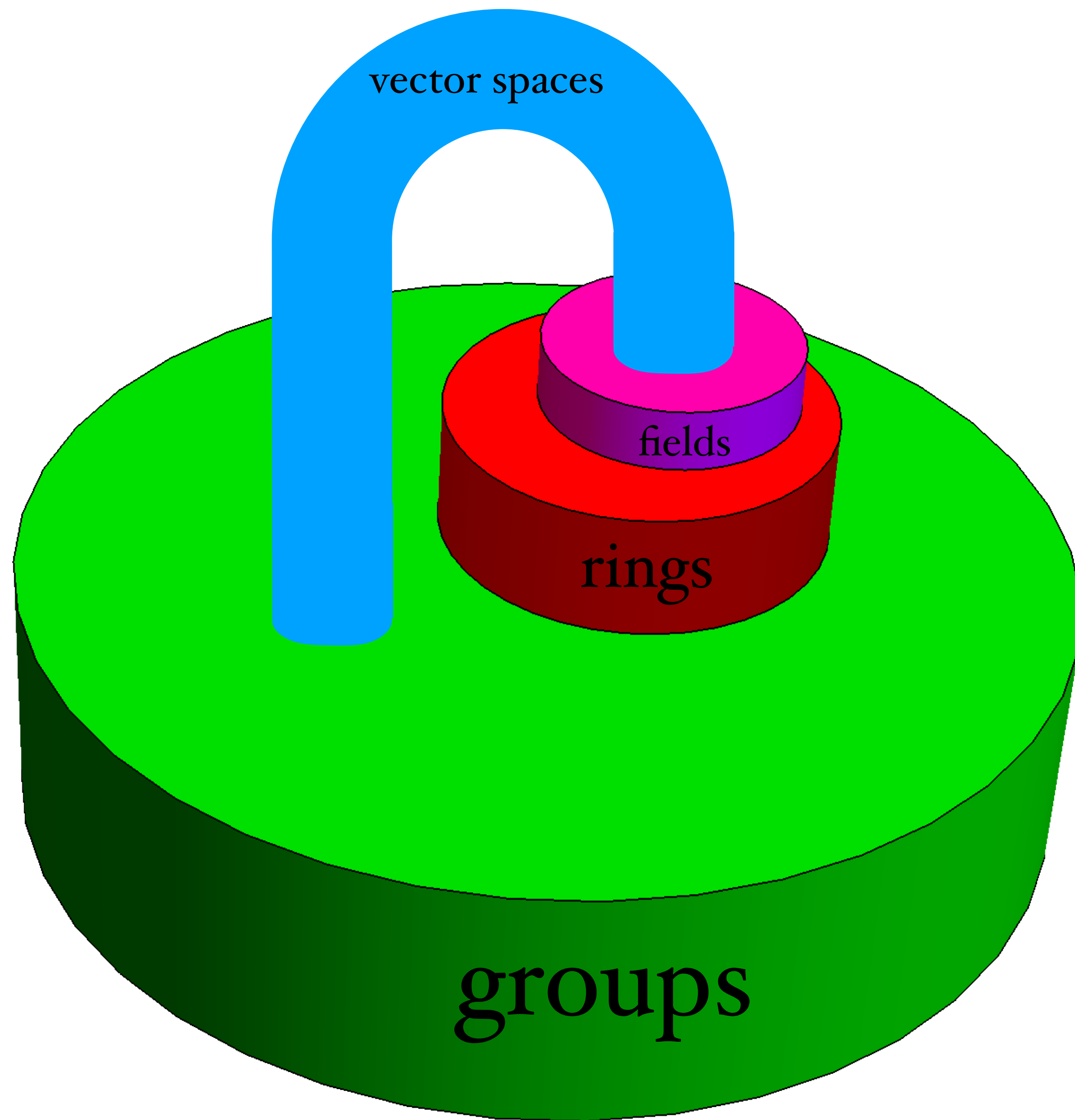
distribution



$+, -, \times, \div$   $\mathbb{Q}$

$+, -, \times$   $\mathbb{Z}$

$+, -$   $\mathbb{Z}$

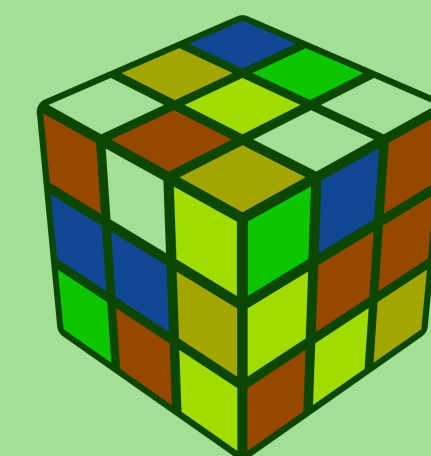


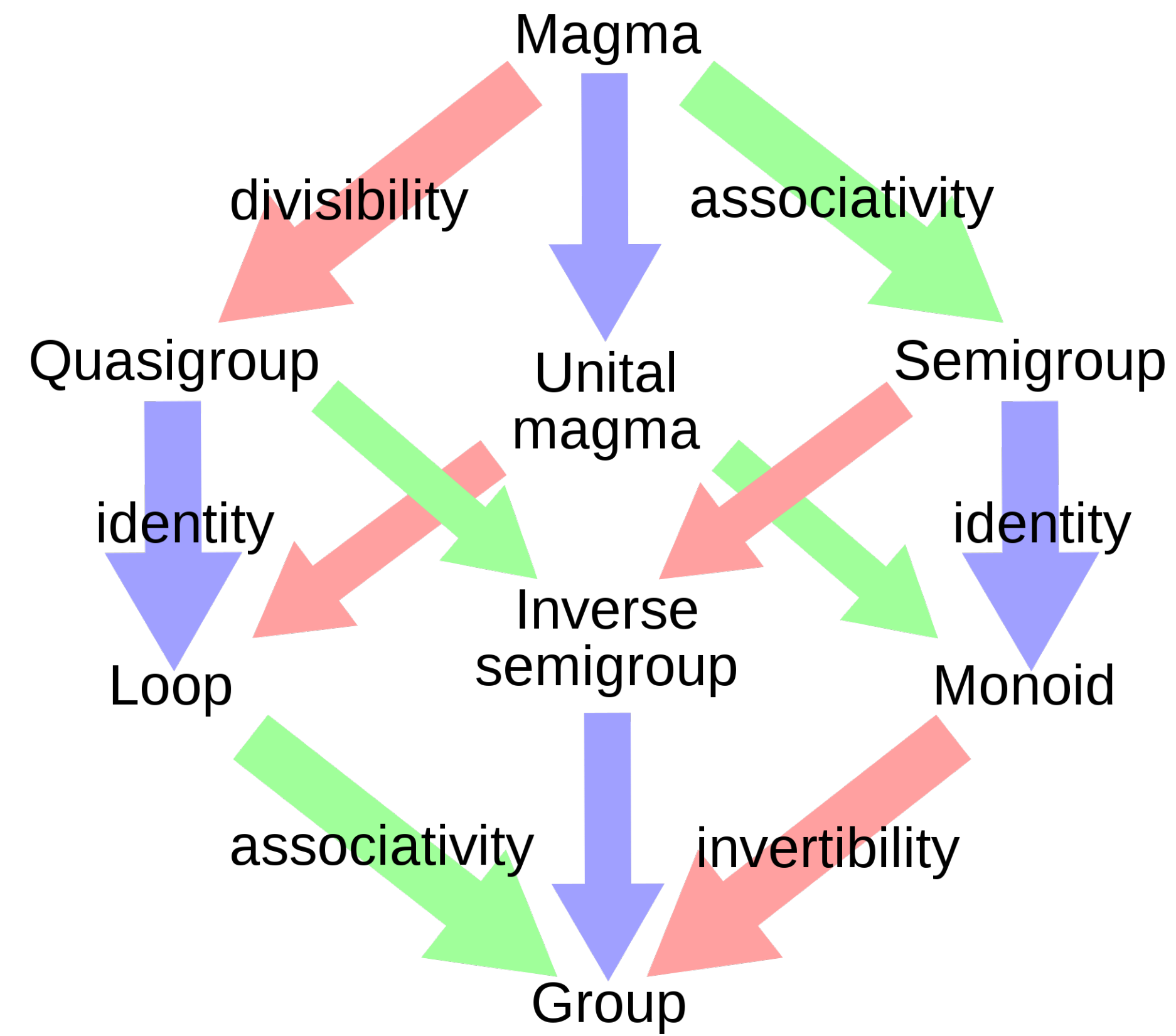
$+, -, \times, \div$   $\mathbb{Q}$  rational functions

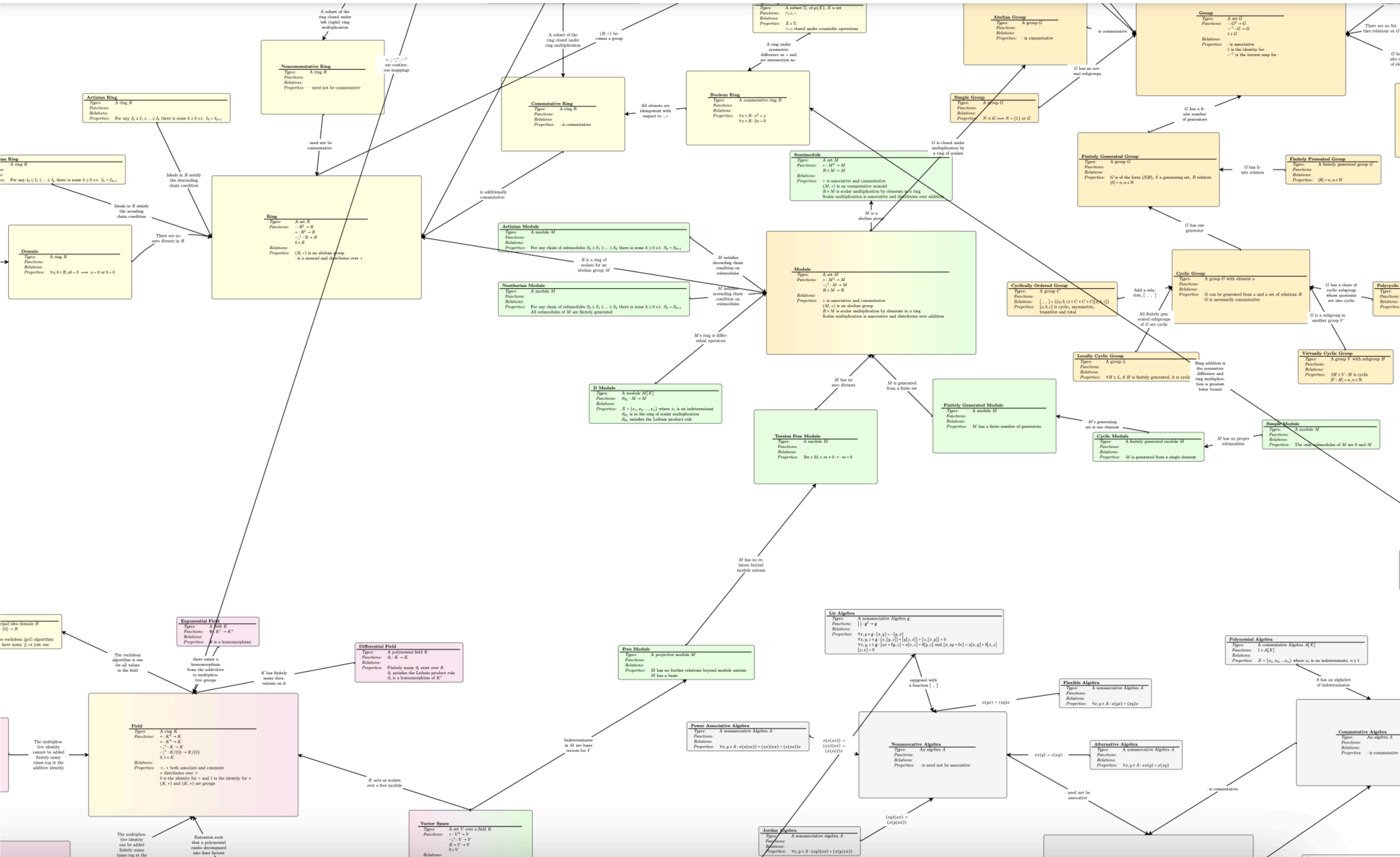
$+, -, \times$   $\mathbb{Z}$  polynomials

$+, -$

$\mathbb{Z}$







$K$  acts as scalars  
over a free module

### Vector Space

*Types:* A set  $V$  over a field  $K$

*Functions:*  $+: V^2 \rightarrow V$

$-_{+}^{-1} : V \rightarrow V$

$K \times V \rightarrow V$

$0 \in V$

*Relations:*

*Properties:*  $+$  is associative and commutative  
 $K \times V$  is scalar multiplication by  $K$   
 $(V, +)$  is a group

# Examples of groups:

$+$ ,  $-$

$\mathbb{R}$  and  $\mathbb{C}$

$\mathbb{Q}$

$\mathbb{Z}/p\mathbb{Z}$

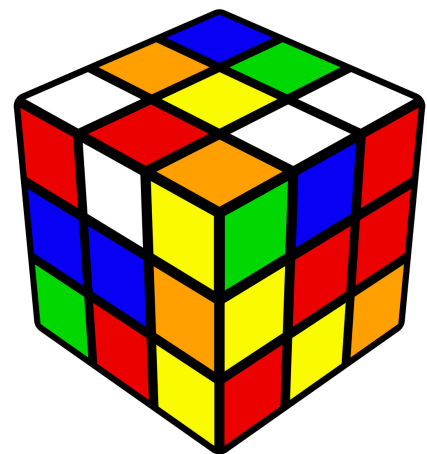
$\mathbb{Z}/6\mathbb{Z}$

ex.  $\mathbb{Z}/5\mathbb{Z}$

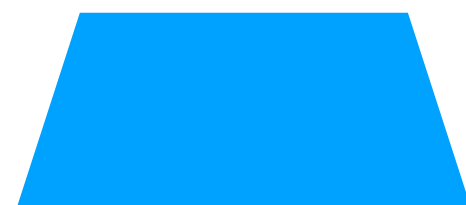
the collection of symmetries  
of any object

$$3 + 4 = 2$$

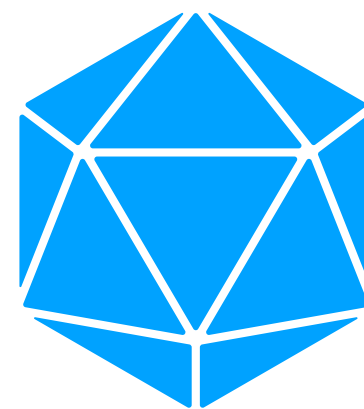
$$3 - 4 = 4$$



$R$



$\mathbb{Z}/2\mathbb{Z}$



$A_5$

# Examples of fields:

$+$ ,  $-$ ,  $\times$ ,  $\div$  (by anything except 0)

$\mathbb{R}$  and  $\mathbb{C}$

$\mathbb{Q}$

$\mathbb{Z}/p\mathbb{Z}$

$\mathbb{Z}/6\mathbb{Z}$



ex.  $\mathbb{Z}/5\mathbb{Z}$

rational functions,  $\mathbb{Z}(x)$  :

$$3x^2 \div \frac{4x^3 + x^2 - 1}{x+1} = \frac{3x^3 + 3x^2}{4x^3 + x^2 - 1}$$

$$3 + 4 = 2$$

$$3 - 4 = 4$$

$$2 \times 4 = 3$$

$$2 \div 4 = 3$$

[https://en.wikipedia.org/wiki/Magma\\_\(algebra\)](https://en.wikipedia.org/wiki/Magma_(algebra))

<http://www.mit.edu/~kepner/GravelMathMap.pdf>